Lecture 15 - Optimization I: The First Derivative Test

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10/29/2026

Updates

Grading taking longer than expected, my apologies!
Review Session: 4:30-6:30 in Vincent Hall 16 (the auditorium in the basement of Vincent).
This place is HUGE so feel free to invite your friends in other sections of 1271 to come.
Today's lecture content will ***NOT*** be tested on Exam 2, it is Exam 3 content.

Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz!

Optimization

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4. Comes from *best quadratic approximation*:
$$\mathscr{Q}_{x=x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 \tag{2}$$

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If $f(c)$ is a global minimum/maximum/extremum then $(c, f(c))$ is a local minimum/maximum/extremum.

Optimization Defns. Example

Let's differentiate (pun maybe intended) the two defns. by examining the wavy function $f(x) = \frac{\sin(x)}{x}$:

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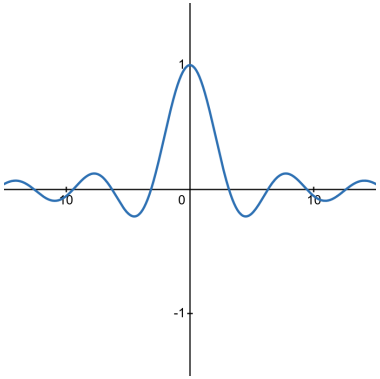


Figure: Graph of $\frac{\sin(x)}{x}$

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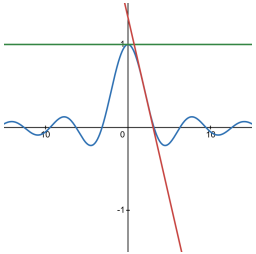


Figure: Graph of $\frac{\sin(x)}{x}$ with two linearizations graphed.

Optimization II

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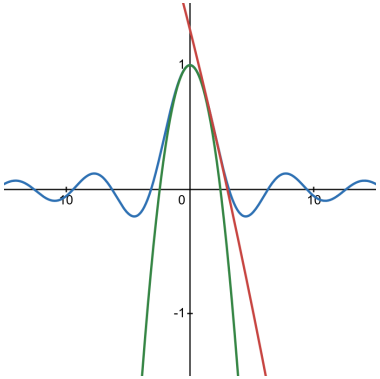


Figure: Graph of $\frac{\sin(x)}{x}$ with two quadratic approximations graphed.

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3. Can we rephrase the condition being a local minimum/maximum in this setting to relate it to conditions on the derivative? Yes!

Optimization IV

Let's assume that $f(x)$ is a function that is both continuously differentiable at $x = x_0$ and that has a local extrema at $(x_0, f(x_0))$.
If we have a local *minimum* at $x = x_0$ then:

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- If $x_0 \leq x < x_0 + \epsilon$ (to the right of x_0) then what can we say?

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- If $x_0 \leq x < x_0 + \epsilon$ (to the right of x_0) then what can we say? Minimality implies $f(x_0) < f(x)$ so by MVT $f'(x) \geq 0$.
- If $f'(x) \leq 0$ for $x < x_0$, $f'(x) \geq 0$ for $x > x_0$ *AND* $f'(x)$ is continuous at $x = x_0$, then that forces $f'(x_0)$

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- If $x_0 - \epsilon < x \leq x_0$ (to the left of x_0) then what can we say about $f(x_0)$, $f(x)$ and $f'(x_0)$? Minimality implies $f(x_0) < f(x)$ and so by mean value theorem, $f'(x) \leq 0$ for $x_0 - \epsilon < x < x_0$.
- If $x_0 \leq x < x_0 + \epsilon$ (to the right of x_0) then what can we say? Minimality implies $f(x_0) < f(x)$ so by MVT $f'(x) \geq 0$.
- If $f'(x) \leq 0$ for $x < x_0$, $f'(x) \geq 0$ for $x > x_0$ *AND* $f'(x)$ is continuous at $x = x_0$, then that forces $f'(x_0) = 0$!

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Moral of the story: At a minimum, we start decreasing (negative derivative) then increase (positive derivative) which forces the derivative at the switching point $f'(x_0)$ to be zero.
Exercise: think about what happens when instead $(x_0, f(x_0))$ is a local *maximum*.

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We can compile this into the following important result:

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Unfortunately, this does not *perfectly* classify local extrema, even for the nicest (infinitely differentiable) functions: consider $f(x) = x^3$:

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This condition is so important we give it its own name:

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The undefined condition is to handle functions like $f(x) = \sqrt[3]{x^2}$:

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- Computing $f(c)$ for every critical point c
- Comparing those above values to find the largest (maximum) and smallest (minimum).

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So we have critical points at $x = 0, 2, -1$.
But are they extrema? I.e. in addition to having $f'(c) = 0$ does $f'(c)$ actually switch signs at $x = c$?

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Thus we've determined that, for $f(x) = \frac{x^5}{5} - \frac{x^4}{4} + \frac{2x^3}{3} + 2$, there is a local minimum at $x = 2$ and a local maximum at $x = -1$.

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Worksheet Time

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!